

Glycine and GABA Pathways in Mouse Retinal Direction-Selective Ganglion Cells

Executive summary

In the mouse retina, the most robust and reproducible pharmacological result is that **GABAergic inhibition is the core null-direction suppressive mechanism for ON-OFF DSGCs**. In the original mouse identification paper, bath-applied **picrotoxin** abolished ON-OFF DSGC direction selectivity almost completely, reducing the direction-selective index (DSI) from 0.68 ± 0.11 to 0.05 ± 0.02 . However, a later subtype-specific mouse study found that a cocktail including **picrotoxin** could leave substantial direction selectivity at **slow** stimulus speeds, with selectivity collapsing only at higher speeds. This is the clearest mouse-to-mouse conflict in the picrotoxin literature, and it strongly suggests that “GABA dependence” is not a single number: it depends on DSGC subtype, stimulus speed, and whether one is measuring spikes or synaptic currents.

1

For **glycine**, the mouse literature points to a different conclusion: glycinergic pathways usually do **not** appear to be the primary source of the classical null-direction signal in DSGCs. Instead, glycine shapes DSGC computation through at least two distinct routes. One route is **serial glycinergic inhibition onto ON starburst amacrine cells (SACs)**, which helps preserve SAC output direction selectivity during sustained grating stimulation; blocking glycine receptors with **strychnine** weakens directionally selective inhibitory input measured in Hb9-positive ON DSGCs. The other route is **direct feedforward glycinergic inhibition onto ON DSGCs**, likely from VGluT3 amacrine cells, which suppresses responses to fast motion and thereby strongly shapes **velocity tuning** with comparatively smaller or even negligible effects on directional tuning under some stimulus conditions.

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The most defensible synthesis is therefore this: **GABA is the principal pathway for null-direction suppression and canonical DSGC direction selectivity, whereas glycine modulates the operating regime of the circuit in a context-dependent way**. Under some conditions, strychnine visibly weakens directionally selective inhibitory currents; under others, it mainly broadens the velocity range over which ON DSGCs respond while leaving directional inhibition largely intact. These differences are not noise; they map onto different glycinergic circuit loci and different stimulus ensembles.

3

Within the primary mouse papers I could verify as directly relevant to the user’s specific drug question, the clearest single-cell electrophysiology studies are: **Weng et al. 2005** and **Trenholm et al. 2011** for **picrotoxin**; **Jain et al. 2022**, **Summers and Feller 2022**, and **Mani et al. 2023** for **strychnine**.

4

Circuit background

Mouse DSGCs comprise multiple classes, but the two most relevant here are **ON-OFF DSGCs** and **ON DSGCs**. ON-OFF DSGCs encode motion over a broad range of speeds, whereas ON DSGCs are specialized for the slow global motion signals that help drive optokinetic image stabilization. In both classes, the canonical

circuit places **starburst amacrine cells** at the center: SAC dendrites are themselves directionally biased and provide **GABAergic** inhibition to DSGCs with strong null-side asymmetry. This framework is well supported by mouse electrophysiology, anatomy, and genetic perturbation studies. 5

The role of **GABAergic inhibition** in mouse DSGCs is therefore conceptually straightforward: it is the most direct and best-established substrate for null-direction suppression. Early mouse single-cell work found that blocking GABA receptors with picrotoxin or bicuculline abolishes ON-OFF DSGC direction selectivity, and later mouse work showed that blocking GABA_A receptors removes the apparent direction selectivity of excitatory currents recorded in DSGCs, reinforcing the idea that **SAC-mediated inhibition is the dominant direction-selective synaptic signal**. 6

The role of **glycinergic inhibition** is subtler. A major mouse finding of the last several years is that glycine acts at more than one locus in the DS circuit. One glycinergic pathway targets **ON SACs**, where narrow-field glycinergic amacrine cells provide slow inhibition that controls SAC gain and, under grating conditions, helps maintain SAC output direction selectivity. A second glycinergic pathway targets **ON DSGCs directly**, where strong feedforward inhibition suppresses responses to fast retinal slip and thereby imposes slow-speed preference. These two pathways predict different outcomes for strychnine depending on stimulus class and readout. 7

Primary studies with explicit strychnine or picrotoxin data

Direct single-cell electrophysiology papers

Primary paper	Species and strain	Preparation	Recording method	Drug(s) and concentrations	Stimulus and DS metric	Key quantitative result
Weng S, Sun W, He S. <i>Identification of ON-OFF direction-selective ganglion cells in the mouse retina.</i> J Physiol. 2005;562(Pt 3): 915-923. DOI: 10.1113/jphysiol.2004.076695. PMCID: PMC1665532.	Mouse, C57BL/6N	Ex vivo isolated whole-mount retina	Loose-patch spike recording; some whole-cell current clamp and voltage clamp	Picrotoxin 100 μM; bicuculline 10 μM	Moving rectangle in 12 directions; DSI	Picrotoxin increased response amplitude and abolished DS, reducing DSI from 0.68 ± 0.11 to 0.05 ± 0.02; 5 cells entered excitatory block. In voltage clamp, excitatory currents were $\sim 20\%$ stronger in preferred direction, whereas inhibitory currents were 3-5$\times$ stronger in null direction.

Primary paper	Species and strain	Preparation	Recording method	Drug(s) and concentrations	Stimulus and DS metric	Key quantitative result
Trenholm S, Johnson K, Li X, Smith RG, Awatramani GB. <i>Parallel Mechanisms Encode Direction in the Retina</i>. Neuron. 2011;71:683–694. DOI: 10.1016/j.neuron.2011.06.020. PMCID: PMC3269126.	Mouse; genetically specified anterior-preferring ON-OFF DSGC subtype; strain unspecified	Ex vivo retina	Cell-attached spikes followed by whole-cell voltage clamp in same cells	Picrotoxin + TPMPA + d-tubocurarine; concentrations unspecified in snippet	Moving spots/bars; DSI from ON and OFF spike responses as function of velocity	Under blocker cocktail, DS persisted at slow speed ($200\ \mu\text{m/s}$); as speed increased, DS collapsed. At $1000\ \mu\text{m/s}$: ON DSI $0.50 \pm 0.08 \rightarrow 0.05 \pm 0.02$; OFF DSI $0.57 \pm 0.07 \rightarrow 0.06 \pm 0.09$; no DS above $1000\ \mu\text{m/s}$.

Primary paper	Species and strain	Preparation	Recording method	Drug(s) and concentrations	Stimulus and DS metric	Key quantitative result
Jain V, Hanson L, Sethuramanujam S, et al. <i>Gain control by sparse, ultra-slow glycinergic synapses</i> . Cell Rep. 2022;38(8):110410. DOI: 10.1016/j.celrep.2022.110410. PMCID: PMC8972185.	Mouse; DSGC experiments in Hb9-GFP superior-motion ON DSGCs	Ex vivo retina	Whole-cell voltage clamp of DSGC IPSCs; also 2-photon imaging of SAC output proxies	Strychnine 1 μM	Drifting gratings, SF 0.1 or 0.01 cycles/deg, TF 1 Hz; DSI of inhibitory currents (PD – ND)/(PD + ND)	At low SF (0.01 c/deg), strychnine increased both null and preferred ON IPSCs: null $25 \pm 10 \rightarrow 59 \pm 12$ pA; preferred $8 \pm 4 \rightarrow 34 \pm 7$ pA; inhibitory-current DSI fell from 0.73 ± 0.08 to 0.28 ± 0.07 ($p < 0.005$, $n = 11$). At high SF (0.1 c/deg), DSI fell from 0.86 ± 0.07 to 0.58 ± 0.07 ($p < 0.005$). OFF IPSCs were largely unchanged.

Primary paper	Species and strain	Preparation	Recording method	Drug(s) and concentrations	Stimulus and DS metric	Key quantitative result
Summers MT, Feller MB. <i>Distinct Inhibitory Pathways Control Velocity and Directional Tuning in the Mouse Retina</i>. Curr Biol. 2022;32(10):2130–2143.e3. DOI: 10.1016/j.cub.2022.03.054. PMID: PMC9133153.	Mouse; Hoxd10-GFP, Drd4-GFP, Trhr-GFP	Ex vivo retina	2-photon targeted whole-cell current clamp and voltage clamp	Strychnine 2 μM	Elongated drifting bars across velocities; DS index for spikes, and decomposition of IPSCs into symmetric versus directional components	In ON DSGCs, strychnine did not measurably reduce directional inhibition, but selectively removed the velocity-tuned symmetric inhibitory component. Symmetric inhibition speed index decreased significantly ($p = 0.04$, 11 cells, 8 mice). Current-clamp recordings were consistent with increased firing but stable directional selectivity at high velocities. No clear strychnine effect on ON-OFF DSGC IPSCs.

Primary paper	Species and strain	Preparation	Recording method	Drug(s) and concentrations	Stimulus and DS metric	Key quantitative result
Mani A, et al. A circuit suppressing retinal drive to the optokinetic system during fast image motion. Nature Communications. 2023. DOI: 10.1038/s41467-023-40527-z. PMID: PMC10447436.	Mouse; ON DSGCs	Ex vivo retina	Whole-cell voltage clamp and spike recordings; optogenetic circuit interrogation	Strychnine; SR95531; concentrations unspecified in retrieved snippet	Fast drifting gratings varying speed; inhibitory charge transfer and spike-speed tuning	Strychnine reduced fast-motion inhibitory charge transfer by $57 \pm 4\%$ ($p = 2 \times 10^{-5}$, $n = 3$); SR95531 alone reduced it by only $17 \pm 7\%$ ($p = 0.08$, $n = 3$); both together abolished inhibition. Strychnine broadened ON-DSGC speed tuning, shifting the half-maximal speed by $1500 \pm 500 \mu\text{m/s}$ ($p = 0.018$, $n = 4$). The paper emphasizes speed tuning more than DSI per se.

Supporting mouse studies that sharpen interpretation

Supporting paper	Why it matters	Main finding relevant to glycine/GABA pathways	Source
Park SJH, Kim I-J, Looger LL, Demb JB, Borghuis BG. <i>Excitatory Synaptic Inputs to Mouse On-Off Direction-Selective Retinal Ganglion Cells Lack Direction Tuning.</i> J Neurosci. 2014. DOI not retrieved in snippet. PMCID: PMC3951696.	Clarifies what GABA blockade does to DSGC synaptic measurements	In whole-cell recordings, null-direction inhibition exceeded preferred-direction inhibition by 2.43 ± 0.31 nS, while excitation had only a small preferred bias (0.31 ± 0.05 nS). After gabazine 50 μM , apparent DS in excitatory conductance vanished ($P - N = -0.0065 \pm 0.039$ nS), supporting the view that SAC GABA is the major DS input and that apparent DS excitation can be a voltage-clamp artifact.	13
Sethuramanujam S, McLaughlin AJ, de Rosenroll G, et al. <i>A Central Role for Mixed Acetylcholine/GABA Transmission in Direction Coding in the Retina.</i> Neuron. 2016. DOI: 10.1016/j.neuron.2016.04.041.	Important background for the GABA/ACh arm of the circuit	In adult mouse DSGCs, the paper argues that SAC-derived ACh and GABA together can dominate low-contrast directional signaling; SAC network stimulation in pharmacological isolation can drive DS responses. This does not center on picrotoxin/strychnine, but it helps localize where GABAergic DS signals enter the circuit.	14

Key figure panels to inspect

The most directly useful figure panels for the user's question are these:

- **Weng 2005, Fig. 2A:** picrotoxin abolishes ON-OFF DSGC spike direction selectivity; the supplementary figure is also important because some cells go into excitatory block. ¹⁵
- **Trenholm 2011, Figs. 4 and 5:** blocker-resistant DS at slow speed, then collapse of DSI at higher velocity in a specific ON-OFF DSGC subtype. ¹⁶
- **Jain 2022, Fig. 5B–E:** strychnine weakens directionally selective inhibitory output from the SAC pathway onto Hb9 ON DSGCs, especially for low-spatial-frequency gratings. ¹⁷
- **Summers 2022, Fig. 7:** strychnine removes velocity-tuned symmetric inhibition while preserving directional inhibition in ON DSGCs tested with elongated bars. ¹⁸
- **Mani 2023, Fig. 1e–h:** strychnine strongly reduces fast-motion inhibition and broadens ON-DSGC speed tuning; SR95531 alone is comparatively modest. ¹⁹

Cross-study comparison

What picrotoxin says about the GABA pathway

The picrotoxin literature in mouse DSGCs is **not monolithic**. In the broad early C57BL/6N ON-OFF DSGC sample of Weng et al., picrotoxin nearly erased DS spiking altogether, strongly supporting the classic “asymmetric inhibition is essential” view. In contrast, the later Neuron paper by Trenholm et al. found that a blocker cocktail containing picrotoxin still permitted robust low-speed DS in a genetically specified ON-OFF DSGC subtype, even though the same responses became nearly nonselective at higher speeds. As a result, the cleanest conclusion is not “picrotoxin either works or fails,” but rather that **GABAergic inhibition is indispensable for classical DS across much of physiological space, while some DSGC subtypes retain a low-speed, inhibition-resistant directional bias that becomes obvious only under particular stimulus conditions.** ¹

The Park et al. study helps explain why these results can coexist. It shows that once GABA_A receptors are blocked, the apparent direction selectivity of excitatory currents disappears, implying that much of what looks like DS excitation in voltage clamp depends on the intact inhibitory context. In other words, one should be careful not to interpret residual DS under GABA blockade as proof that inhibition is unimportant; it may instead expose **postsynaptic nonlinearities or dendritic asymmetries** that are normally layered on top of SAC-mediated inhibition. ²⁰

What strychnine says about the glycine pathway

The strychnine literature in mouse is even more context dependent. Jain et al. found that strychnine **weakens the direction selectivity of inhibitory currents** arriving at Hb9 ON DSGCs, with particularly strong effects for **low-spatial-frequency drifting gratings**. Under those conditions, glycinergic inhibition appears to act upstream of DSGCs by stabilizing SAC output direction selectivity during sustained, spatially broad stimulation. ¹⁷

By contrast, Summers and Feller used **elongated drifting bars** and decomposed ON-DSGC inhibition into a directional component and a symmetric component. Under that assay, strychnine removed the **velocity-tuned symmetric inhibition** while leaving the **directional inhibitory component** essentially intact. This implies that the glycinergic pathway engaged by elongated bars is largely **not** the same pathway that generates the canonical null-direction inhibition; instead it shapes how strongly ON DSGCs are suppressed as stimulus speed rises. ²¹

Mani et al. are strongly consistent with Summers on this point. Their fast-motion study found that strychnine removed more than half of the inhibition evoked by rapid grating motion and strongly broadened the speed range over which ON DSGCs could fire, whereas a pure GABA_A blocker alone had only a small effect. Again, the dominant story is **speed tuning**, not a wholesale loss of directional coding. ¹⁹

Interpreting the conflicts

The apparently conflicting picrotoxin and strychnine results become coherent when the circuit locus, stimulus regime, and measured variable are separated.

First, the **stimulus ensemble** matters enormously. Jain et al. explicitly argue that slow glycinergic inhibition of SACs is best recruited by **drifting gratings**, especially large low-spatial-frequency gratings, whereas classic moving spots or annular stimuli can under-recruit that pathway. That provides a direct explanation for why some studies saw strong strychnine effects on SAC output DS while others saw little change in DSGC DS. ²²

Second, the **readout** matters. Weng measured spiking DSI. Trenholm examined spiking and synaptic currents in a subtype with strong dendritic asymmetry. Park asked whether excitatory conductances were themselves direction selective. Jain measured DSGC inhibitory currents as a proxy for SAC output directionality. Summers decomposed ON-DSGC inhibition into directional versus symmetric components. Mani focused on inhibitory charge and spike-speed tuning during fast motion. These are not interchangeable observables. A drug can weaken **directional inhibition**, leave **spike DSI** relatively intact, or strongly change **speed tuning** without producing the same headline conclusion. ²³

Third, **cell type and subtype** matter. The picrotoxin-resistant low-speed DS described by Trenholm et al. was observed in a **genetically specified anterior-preferring ON-OFF DSGC subtype**, not in a broad survey of ON-OFF DSGCs. Likewise, the strongest mouse strychnine results concern **ON DSGCs**, especially Hb9-positive or related optokinetic-projecting types, not ON-OFF DSGCs in general. ²⁴

Fourth, **drug specificity and network spread** complicate interpretation. Picrotoxin is broader than a selective GABA_A antagonist because it also blocks GABA_C/ρ-type receptors; some studies additionally combined it with TPMPA and d-tubocurarine. Strychnine is usually used at 1–2 μM in the mouse DSGC literature cited here, but the broader literature has pointed out that micromolar strychnine can have non-glycinergic effects in some retinal contexts. These pharmacological caveats do not invalidate the conclusions, but they counsel against over-reading any single blocker experiment as a perfectly pathway-isolated test. ²⁵

Gaps and follow-up experiments

The most important unresolved issue is that the mouse literature still lacks a **single unified within-preparation experiment** that compares **picrotoxin-versus-gabazine** and **strychnine** in the **same identified DSGC subtype**, while using **both elongated bars and drifting gratings**, and while measuring **spikes, EPSCs, IPSCs, and upstream SAC signals**. That experiment would directly test whether the Weng, Jain, Summers, and Mani results are different because of pathway locus, subtype, or stimulus statistics. ²⁶

A second gap is pathway assignment. The glycinergic effects seen by Jain et al. are most consistent with **narrow-field amacrine cell** → **ON SAC** inhibition, whereas the effects seen by Summers and Mani are most consistent with **VGlut3 amacrine cell** → **ON DSGC** inhibition. The decisive experiment would be to combine **cell-type-specific synaptic silencing or transmitter-release knockouts** for VGlut3 cells and narrow-field glycinergic amacrine cells with the same electrophysiological assays used in those papers. ²⁷

A third gap concerns the residual low-speed DS under GABA blockade seen by Trenholm et al. The field still needs a direct replication in modern mouse lines with visual stimuli matched to Weng et al. and readouts matched to Park et al. If the same subtype-dependent residual DS is reproduced, then the cleanest current model would be that **SAC GABA is the dominant DS input, but dendritic or spike-threshold**

nonlinearities can preserve weak DS output when inhibition is compromised, especially at slow speeds.

28

A fourth gap is developmental and state dependence. The exact-drug mouse papers identified here are overwhelmingly in mature ex vivo retina, and they do not systematically compare dark adaptation, luminance regime, or developmental age. Because glycinergic pathways are especially state dependent in the inner retina, repeating the same blocker assays across adaptation states and ages would likely explain part of the current heterogeneity. 29

Overall, the most rigorous current view is this: **GABAergic SAC input is the principal engine of mouse DSGC direction selectivity, whereas glycinergic pathways tune when and how strongly that direction-selective circuit operates.** Under broad sustained gratings, strychnine can weaken SAC-derived directional inhibition; under elongated bars and fast image motion, it mainly lifts a glycinergic brake on ON-DSGC firing and shifts speed tuning. The seeming contradictions are therefore better understood as signatures of **parallel inhibitory pathways** rather than mutually exclusive models. 30

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9 16 24 28 **Parallel Mechanisms Encode Direction in the Retina - PMC**
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